HOMOTOPY COMMUTATIVE SQUARES INDUCE MAPS OF CONES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2736-2753, following the statement at 2717-2734.

Editorial note. No omissions: the homotopy data and the matrix for c are displayed; the differential check is a mechanical 2x2 block computation and the i/p compatibilities are read off matrix blocks.

Allowed standard prerequisites (editorial).

- Mapping cone and its canonical maps. definition-cone (same chapter, at ~2688). Supplies C(f) on L^n + K^{n+1} with i and p.

Lemma 0.1. *Suppose that*

$$\begin{array}{ccc} K_1^\bullet & \xrightarrow{f_1} & L_1^\bullet \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ K_2^\bullet & \xrightarrow{f_2} & L_2^\bullet \end{array}$$

is a diagram of morphisms of complexes which is commutative up to homotopy. Then there exists a morphism $c : C(f_1)^\bullet \to C(f_2)^\bullet$ which gives rise to a morphism of triangles $(a, b, c) : (K_1^\bullet, L_1^\bullet, C(f_1)^\bullet, f_1, i_1, p_1) \to (K_2^\bullet, L_2^\bullet, C(f_2)^\bullet, f_2, i_2, p_2)$ of $K(\mathcal{A})$.

Proof. Let $h^n : K_1^n \to L_2^{n-1}$ be a family of morphisms such that $b \circ f_1 - f_2 \circ a = d \circ h + h \circ d$. Define c^n by the matrix

$$c^n = \begin{pmatrix} b^n & h^{n+1} \\ 0 & a^{n+1} \end{pmatrix} : L_1^n \oplus K_1^{n+1} \to L_2^n \oplus K_2^{n+1}$$

A matrix computation show that c is a morphism of complexes. It is trivial that $c \circ i_1 = i_2 \circ b$, and it is trivial also to check that $p_2 \circ c = a \circ p_1$. □

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